

B.Sc. (Entire Computer Science)-I, Level - 4.5 UG Certificate Level							
Sem:		II					
Paper Category:		GE2/OE2					
Paper Name:		Software Engineering			Paper Code : G06-GE-OE-201		
Credit:		02		Theory:		2 Hrs./Week	
Marks:	UA:	30	CA:	20	Total:	50	

Course Objectives –

The aim of this course is to prepare learners for a foundational understanding of computers, encompassing:

1. To understand the fundamental concepts and characteristics of systems.
2. To categorize different types of systems and understand system analysis.
3. To explore various SDLC models and their applications with different functional and non-functional requirements.
4. To learn various techniques for gathering system requirements.
5. To understand the process of designing and implementing a system.
6. To understand coding standards, size measures, complexity analysis, and verification.
7. To grasp the fundamentals of software testing.
8. To learn about software implementation and the maintenance process.

Course Outcomes-

1. Students will be able to define a system, identify its characteristics, and describe various types of systems.
2. Students will be able to describe and compare models such as the Waterfall model, V-shape model, Spiral model, and Prototyping, Incremental, RAD, and Agile methodologies.
3. Students will be able to identify, document, and analyze user and system requirements.
4. Students will be proficient in conducting interviews, questionnaires, record reviews, and observations for requirement gathering.
5. Students will be able to design data flow diagrams, entity-relationship diagrams, structured charts, and create a data dictionary. They will also learn input and output design.

6. Students will learn to write clean, maintainable code, estimate effort and cost, and verify code correctness.
7. Students will be able to perform different types of testing and understand testing methodologies.
8. Students will understand the steps involved in implementing software and the different types of software maintenance.

Unit I: - Introduction to Software Engineering:	No. of Lectures:15	Weightage: 12-18 Marks
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System concepts: Introduction system, characteristics, Elements of system, Types of system, System Analysis, Role of System Analyst, Software Engineering: Definition, Characteristics of software, Qualities of software. System Development life cycle: Waterfall model, V-shape model, Spiral model, Prototyping, incremental, RAD, Agile. Software requirements: Functional, Non-functional requirements, User requirements, System requirements, Fact-finding techniques: Interviews, Questionnaires, Record reviews, Observation Analysis and Design Tools: Flowcharting, Decision tables, Decision Trees, Structured English, Structure charting Techniques.

Unit II: - System Design and Implementation, Maintenance:	No. of Lectures:15	Weightage: 12-18 Marks
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Data flow Diagram (Physical, Logical), Entity relation diagram, structured chart, Data Dictionary, Input and output design

Coding: Verification, size measures, complexity analysis, coding standards, Effort Estimation, Cost Estimation, Testing and its types, need

Construction of the system: traditional and incremental approaches, conversion methods, Software Implementation, Overview of the maintenance process, and types of maintenance.

Reference Books-

1. Analysis and Design of Information Systems By James Senn.
2. Practical guide to structure System Design By Miller/Page/jones.
3. Software Engineering By Pressman.
4. System Analysis and Design By Parthsarty